d 12

$$l = y \log(1+e^{-t}) + (1-y)\log(1+e^{t})$$

$$\frac{\partial l}{\partial \theta_j} = \frac{\partial l}{\partial t} \cdot \frac{\partial t}{\partial \theta_j} \quad ; \quad t = \theta^T x$$

(logit)

$$\frac{\partial l}{\partial t} = -y \frac{e^{-t}}{1+e^{-t}} + (1-y)\frac{e^{t}}{1+e^{t}}$$

$$= -y \frac{e^{t}}{1+e^{t}} \cdot e^{-t} + (1-y)\frac{e^{t}}{1+e^{t}}$$

$$= \frac{-y + e^{t} - ye^{t}}{1+e^{t}}$$

$$= \frac{e^{t}}{1+e^{t}} - y\left(\frac{1+e^{t}}{1+e^{t}}\right)$$

$$= g(t) - y$$

$$\therefore \quad \frac{\partial l}{\partial \theta_j} = \left(g(t) - y\right) \cdot x_j$$

(log - likelihood)

$$l = y \log h(x) + (1-y) \log(1 - h(x))$$

$$l_{CE} = -l = -y \log \frac{1}{1+e^{-t}} + (y-1) \log\left(1 - \frac{1}{1+e^{-t}}\right)$$

$$= y \log(1+e^{-t}) + (y-1) \log\left(\frac{1+e^{-t}-1}{1+e^{-t}}\right)$$

$$= y \log(1+e^{-t}) + (y-1) \log\left(\frac{e^{-t} \cdot e^{t}}{1+e^{-t} \cdot e^{t}}\right)$$

$$= y \log(1+e^{-t}) + (y-1) \log\left(\frac{1}{1+e^{t}}\right)$$

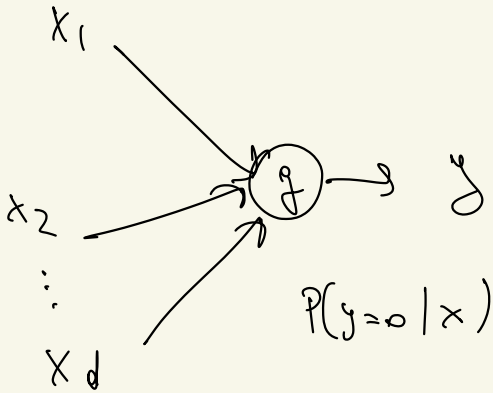
$$= y \log(1+e^{-t}) + (1-y) \log(1+e^{t})$$

$$l_{CE} = -\log(\phi_y) = -\log\left(\frac{e^{t_y}}{\sum_{j=1}^{K} e^{t_j}}\right)$$

$$l_{CE} = -t_y + \log\left(\sum_j e^{t_j}\right)$$

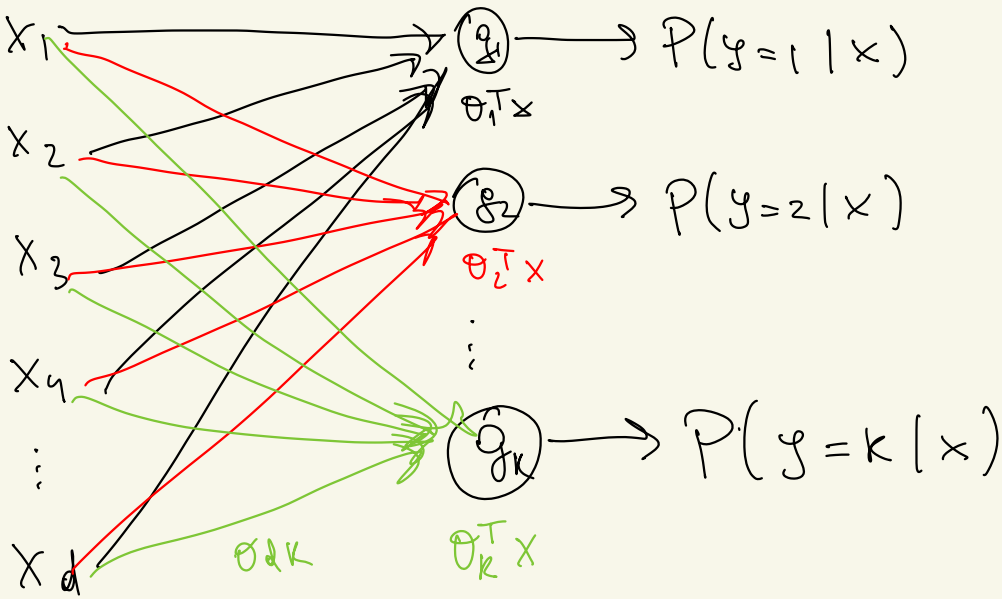
$$\frac{\partial l_{CE}}{\partial t_i} = -1\{y=i\} + \underbrace{\frac{1}{\sum_j e^{t_j}} \cdot e^{t_i}}_{\phi_i}$$

$$= \phi_i - 1\{y=i\}$$



$$P(y=1|x)$$

$$P(y=0|x) = 1 - P(y=1|x)$$



$$P(y=1|x)$$

$$P(y=2|x)$$

$$P(y=k|x)$$